A HIERARCHICAL ALGORITHM FOR NEARFIELD ACOUSTIC IMAGING

Michael Peake[†] *Mehmet Karan*[†] *Doug Gray*[†,‡]

[†] Cooperative Research Centre for Sensor Signal and Information Processing (CSSIP)
1 Warrendi Rd. Mawson Lakes, SA 5095, Australia
[‡] Department of Electrical and Electronic Engineering, University of Adelaide, Australia
mpeake@cssip.edu.au mehmet@cssip.edu.au dgray@cssip.edu.au

ABSTRACT

This paper presents a computationally efficient hierarchical nearfield imaging algorithm using a delay and sum beamformer with a random array via pulse-echo techniques.

In the conventional imaging of nearfield sources, signals from multiple sensors are delayed and summed to focus at points in the imaging volume. However, the computational burden of implementing the conventional algorithm for arrays with a large number of receivers, due to the distance calculations, requires new suboptimal algorithms. The algorithm described in this paper reduced computational time balanced against a reasonable memory requirements. The key principle of this algorithm is to implement the delay and sum by grouping receivers into various subarrays in a number of stages. Mirroring the hierarchy of subarrays is a decomposition of imaging volume beginning with large voxels and progressively reducing the voxel size. This algorithm introduces two main changes to the conventional algorithm which are *(i)* subsampling of the received data, which is then compensated by phase interpolations, *(ii)* a hierarchy of subarrays and subvolumes to combine the data from all the receivers.

Comparison of the conventional and the hierarchical algorithms are done in terms of their point spread functions.

1. INTRODUCTION

In this paper, we consider the problem of nearfield imaging using an acoustic transmitter with a bandwidth of several MHz and a receiving array of several thousand randomly positioned sensors. Acoustic imaging has found applications in diverse areas such as medical diagnosis, oceanic search, non-destructive evaluation and exploration seismology [1]. When the object is assumed to be in the farfield of an array, the received waveform from a single point reflector can be assumed to be planar. Farfield approximation usually starts around at a range $r = L^2/\lambda$ as described in [2] where r is the distance from the array, L is the diameter of the array and λ is the wavelength of the transmitted signal. However, when the object to be imaged is within this distance, new techniques such as in [3] need to be developed.

In underwater acoustic imaging, a common technique is to use a delay and sum beamformer. This technique obtains the image of an object from the delayed and summed backscattered echoes of a pulse modulated signal emitted by a transmitter. A carrier frequency in the region of 3-5 MHz is a compromise between low angular resolution at the lower frequency and high absorption at higher frequencies. As discussed in [4], use of a linear FM signal or "chirp" allows lower power transmission for a given range resolution and a bandwidth of several MHz allows millimetre resolution to be achieved. To obtain good cross-range imaging, a large receiver aperture array is required - typically of the order of 1 m^2. The number of sensors required to densely fill such an aperture is prohibitive and thus a sparse array must be considered. The sensor locations are chosen randomly to reduce grating lobes that may otherwise occur (see [5, 6] and the references therein).

The computational burden of conventional time delay and sum imageformers using outputs of all the receivers is high [7]. Two features conspire against us. Firstly, the sparseness of the array means that subarrays have extremely directional beam-patterns and high sidelobes. If we combine receivers into a subarray by delaying and summing their data streams, that sum will only generate a good quality image of a small patch. Thus, this process must be repeated for each small patch to cover the full image. The second feature that makes the task difficult is that the object to be imaged is in the nearfield. Approximations that can be used in the farfield do not all apply here. For ex-

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ample, when calculating the distance from a receiver to many voxels in the image, either a time-consuming formula or a vast lookup table is required.

In acoustic imaging [4], an object is usually modelled in two ways: the object region is composed of a number of either point reflectors or several homogeneous media. In this paper, we employ the first assumption. Therefore the image of the object is assumed to be a superposition of point spread functions of the point reflectors in the object model. Hence, point spread functions are used as quality measures for the acoustic imaging algorithms in this paper.

In this contribution, we propose three novel techniques to address this nearfield acoustic imaging problem.

First, a piecewise polynomial approximation is used to quickly generate the distance calculations on the fly.

Secondly, the nature of the outputs of the matched filter used in the ranging processing is exploited to allow the time delay and sum imageforming to be implemented using coarse time delays followed by a fine time delay adjustment using phase multiplications.

Finally, and the main contribution presented, the imageforming is carried out in a hierarchical manner by grouping receivers into various subarrays in a number of stages. Mirroring the hierarchy of subarrays is a decomposition of the imaging volume beginning with large voxels and progressively reducing the voxel size.

In the next two sections, we describe the conventional and hierarchical algorithms and address some implementation issues.

2. THE CONVENTIONAL NEARFIELD ACOUSTIC IMAGING ALGORITHM

In conventional nearfield imaging [4, 7], a wideband signal, typically a linear FM chirp, is transmitted and backscattered echoes are received by an array of sensors. The sensor outputs are sampled, Hilbert transformed to produce complex signals and "dechirped" by convolving with a complex replica of the transmitted signal. The typical range profile from a single receiver is shown in Figure 1. To image in cross-range at each voxel, the round-trip distance is first calculated from the transmitter via the voxel to each receiver. The corresponding sample is then chosen from the dechirped data received from each sensor. These samples are summed, and the absolute value of the sum is chosen to be the voxel intensity.

This conventional algorithm is not practical due to high computational resource requirements. Therefore, we have altered that algorithm to achieve a balance between the speed, memory size and the image quality.

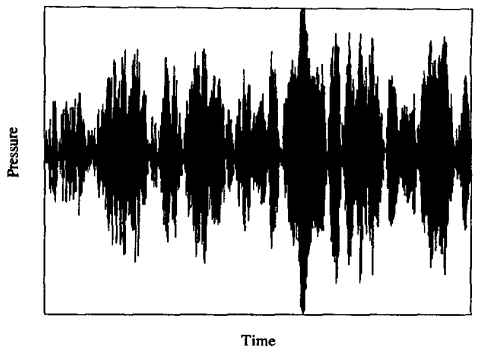


Figure 1: Range profile from a single receiver

The next section describes the new algorithm.

3. A HIERARCHICAL NEARFIELD ACOUSTIC IMAGING ALGORITHM

The hierarchical algorithm proposed in this paper divides the whole array into subarrays, and processes the data in a hierarchical manner. This algorithm is similar to the Quadtree Backprojection algorithm described in [8] for use with impulse radars and the fast beamforming algorithm in [9]. Depending on the number of algorithm stages chosen, the array is first divided into a number of subarrays of roughly equal area. These are divided into smaller arrays until we reach individual receivers. For example, if the algorithm is implemented such that it has three stages, then it divides the array into only small and large subarrays. If n_1 large subarrays are used and each large subarray contains n_2 points which are the centre points of small subarrays, then these $N_2 = n_1 n_2$ points would form the centres of small subarrays. By varying the position and number of large and small subarrays, one can change the performance and the computational complexity of the hierarchical algorithm.

The hierarchical algorithm uses the key idea that a subvolume of many voxels will lie in the farfield of a small subarray, allowing shortcuts in the computation. Smaller subvolumes lie in the farfield of larger subarrays. Thus we deal with either fewer subvolumes and all the array, or all the voxels and a few large subarrays. By combining receivers into small subarrays and then into large subarrays, we can focus on first large image subvolumes and then on smaller ones and finally on voxels. This process is similar to the butterfly structure in FFT or the Butler matrix used in radar. However, in these cases where we are dealing with plane waves the butterfly decomposition is exact, whilst in

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this approach it is approximate. Therefore, the proposed algorithm has a hierarchical structure where the concurrent contraction in the imaging subvolumes and an expansion in array subareas continue with an improvement in image resolution at each stage. This fact allows us to reduce the number of computations at each stage by a significant factor. A pseudo-code of the operations carried out in a stage of the algorithm can be seen in Table 1. In order to further reduce the algo-

```
for each large subarray
  load cubic coeffs for n_i small subarrays
  for each large subvolume
    for each of the n_i small subarrays
       load data stream, apply phase-shifts
       for each small subvolume
            Calculate distance
            Convert to (sample number, phase)
            Add data substream to new substream
    save new substreams
    zero new substreams for next large subvolume
```

Table 1: Pseudo-code of one stage of the hierarchical imaging algorithm

rithm requirements, both in computation and memory, the two techniques described in the following section can be used. These techniques may also be incorporated in the conventional algorithm.

4. IMPLEMENTATION ISSUES

In many software and hardware implementations, excessive delays can be caused by pre-calculating delay values and transferring the required blocks of data and the delays in and out of cache memories. It is often more efficient to calculate time delays on the fly which requires fast algorithms. The following technique can be used to do so and it will be employed in the new hierarchical algorithm described in the next section.

The distance can be approximated well over part of the imaging volume by a cubic polynomial. We will use several polynomials P_i, each covering a part V_i of the imaging volume. The coefficients in each polynomial P_i are found by regression using several distances within V_i. This can be done off-line and only twenty coefficients per cubic polynomial need be stored.

To evaluate each cubic polynomial over many equally spaced voxels, we use the following method. We could evaluate a linear function $f(n) = a + bn$ at integers $n = 0, 1, 2, \ldots$ simply by starting with a and continually adding b to the previous answer. This uses one addition to evaluate $f(n)$ for each value of n. In the same way, by nesting these additions three levels deep we can evaluate a cubic polynomial using just three adds for each value of n.

In addition to distance approximations, extra speed-up of acoustic imaging algorithms may be possible via subsampling and then interpolating received signals. In many applications, design considerations dictate a sampling rate that exceeds the transmitted signal bandwidth by a factor of 4-10. The impulse response of the matched filter output (ie. the range compressed chirp) typically looks like Figure 2. Time delaying a receiver

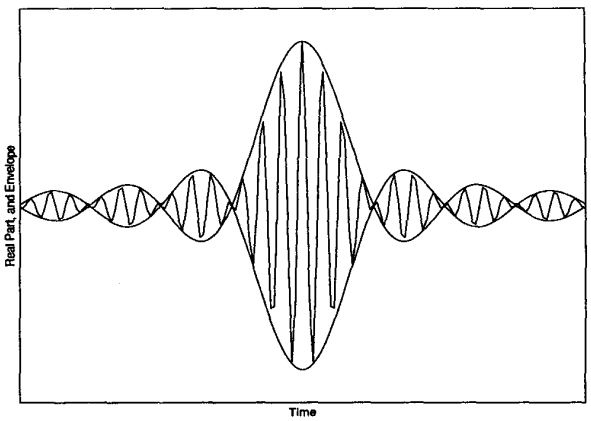


Figure 2: The output of a filter matched to a linear FM signal driven by the same signal.

output can be achieved by first selecting at a suitable subsampling factor the sample nearest to the peak of the modulation - this is termed coarse time delay. Provided we are within the 3 dB point of the peak, a fine time delay adjustment may be made through multiplication by the appropriate phase factor. This subsampling (typically 4-8 in practice) allows substantial savings in down-stream computations as it enables the hierarchical algorithm of the following section to be implemented.

The above techniques can be used also with the conventional algorithm. A combination of these ideas are used in the proposed algorithm to reduce the number of calculations to a small fraction of the number needed by the conventional acoustic imaging algorithm.

5. SIMULATION STUDIES

Figure 3 gives the cross-sections of the point spread functions of the conventional and hierarchical algorithms in the (x, y) plane, while Figure 4 shows similar cross-sections in the (x, z) plane. In this example, an ar-

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ray of around 2500 receivers pseudo-randomly spaced in a (50cm)x(50cm) aperture was used, implying an average separation of about 10λ. A three-stage hierarchical algorithm was employed with 21 large subarrays and sixteen small subarrays in each large subarray. The decomposition of the image volume was 9, 3 and 1mm respectively at a range of one metre. The subarrays, as well as the receivers, formed irregular patterns. We have found that regular subarrays gave rise to high sidelobes, even if the receivers had no regularity at all.

Figure 3 shows a raised sidelobe floor in the region near the mainlobe. The value of the point spread function is much higher in the central subvolume than in the surrounding subvolumes. Also, the position of the point reflector relative to the coarse subvolumes affects the point spread function. Figures 3 and 4 show the result when the point reflector happens to be at the centre of a coarse subvolume, which is when the algorithm does best. Greater errors are introduced by the algorithm when the point reflector is the edge of a coarse subvolume. The cross-section in the range direction can be seen in Figure 4. It can be observed that most of the blurring is in the cross-range direction. The hierarchical algorithm causes sidelobes to blur more in the range direction, while as Figure 3 indicates, the mainlobe is blurred mostly in the cross-range direction.

6. CONCLUSION

In this paper, we have presented a hierarchical method for nearfield acoustic imaging using a large, pseudo-random array of sensors. This algorithm achieves a slightly degraded image with a reduced amount of memory in a shorter computation time. The algorithm applies several modifications to the conventional imaging algorithm. These modifications are as follows:

- The use of a hierarchy of subarrays allows a significant reduction in the number of calculations required. The amount of time saved depends on the resolution required compared with the density of receivers in the array. However, close attention must be paid to the arrangement of subarrays to limit degradations to the image.
- The data was down-sampled to reduce the number of operations required. The resulting image would be severely degraded unless interpolations were applied in the form of complex phase-shifts. We allowed only a quantized set of phase-shifts. These phase-shifts were found efficiently, and did not cause a great degradation in the image.
- The calculation of time-delays, which forms a large part of the conventional algorithm, was achieved in the hierarchical one with polynomial approximations in a few adds each.

Moreover, the hierarchical algorithm allows minimal interaction with the hard disc, which is large enough to contain all the data but slow to fetch data from. This is as important as the reduction in number of calculations.

Point-spread functions were presented to compare the two algorithms. Raised sidelobe floors were seen in the hierarchical algorithm's point spread function. These can be reduced by a judicial choice of irregular subarrays.

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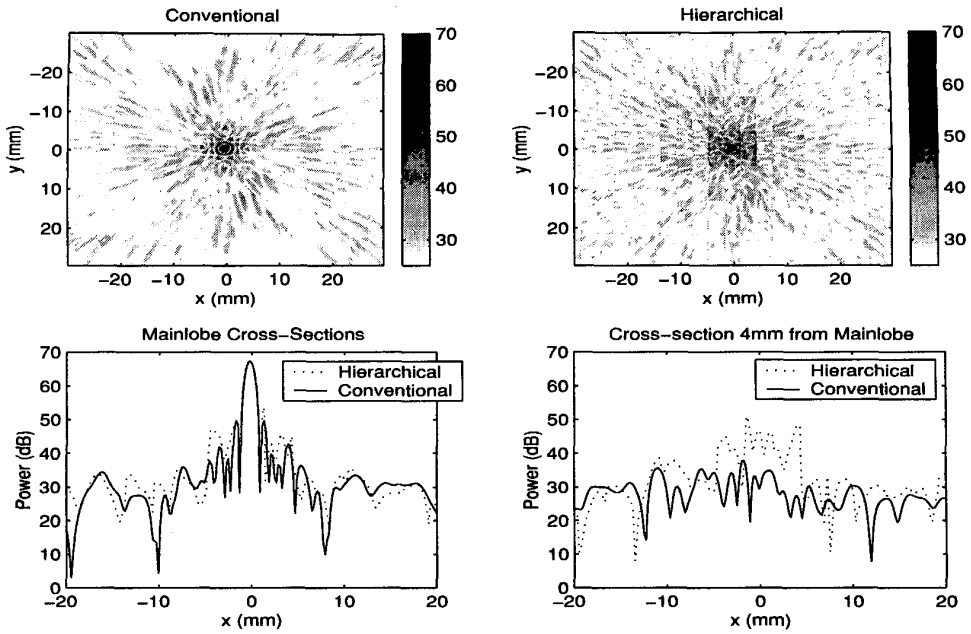


Figure 3: Conventional and hierarchical algorithm point spread functions in the (x, y) plane. Intensities shown in dB scale. The lower graphs show the cross-sections of the point spread functions of hierarchical and conventional algorithms at $y = 0$ and $y = 4$ mm.

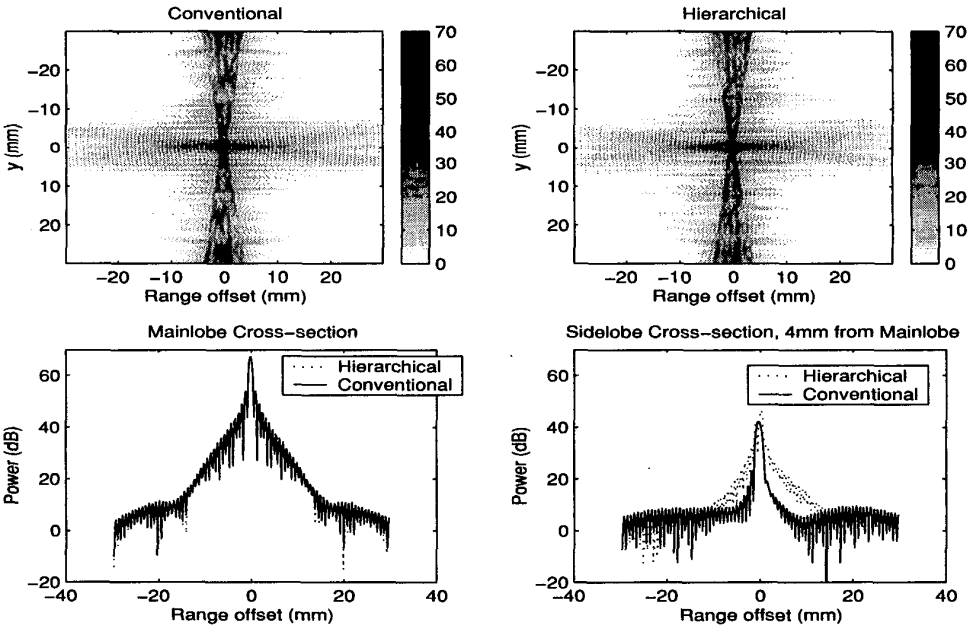


Figure 4: Conventional and hierarchical algorithm point spread functions in the (x, z) plane. Intensities shown in dB scale. The lower graphs show the cross-sections of the point spread functions of hierarchical and conventional algorithms at $y = 0$ and $y = 4$ mm.

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